
Course Information

Course: Foundations of Artificial Intelligence
Credit Hours: 3
Room: SCI 103
Day/Times: Mon, Wed 16:00 P.M.- 17:10 P.M.
Prerequisites: COMP100, MATH107, ENGR200, and MATH203.
Textbook: Christopher M. Bishop with Hugh Bishop (2024).
Deep Learning Foundations and Concepts. Springer.

Instructor Information

Name: Dr. Fatma Guney
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Course description

This course provides a broad introduction to the fundamental principles and techniques of machine learning. Students will learn how machines can automatically extract patterns from data, make predictions, and improve their performance over time. The course covers the basics of supervised and unsupervised learning, core algorithms for regression and classification, optimization and regularization techniques, and the principles of model evaluation. Students will also gain hands-on experience with data preprocessing and representation learning. By the end of the course, students will be able to implement, analyze, and evaluate basic machine learning algorithms and understand their strengths, limitations, and applications.

Topics

- Introduction to learning paradigms
 - Formulation of simple machine learning (ML) as an optimization problem
 - Overfitting/controlling solution complexity with regularization
 - Data processing
 - Principles of ML evaluation
 - Unsupervised learning and clustering
 - Representations from manually crafted to learned
 - Basic neural networks
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Learning Outcomes

By the end of this course, participants will be able to:

- Explain fundamental machine learning paradigms and algorithms
- Formulate and solve machine learning problems as optimization tasks
- Preprocess, represent, and analyze data effectively
- Evaluate and compare machine learning models in a principled manner
- Implement and experiment with core machine learning models

Course Evaluation and Grading Policies

Grades are based on: Quiz (10%), midterm (25%), homework (30%), final exam (35%).

Students are required to attend at least 70% of the scheduled class sessions. Attendance below this threshold will result in automatic failure of the course, regardless of performance in exams or assignments.

There will be only one make-up exam offered during the semester. The make-up exam is comprehensive and will cover the material from all exams. Students who miss more than one exam will not be eligible to take multiple make-ups; the single make-up exam will be the only option provided.

Topic Schedule

Week 1		Introduction to Learning Paradigms
Week 2		Machine Learning Mechanics – Terminology and Techniques
Week 3		K-Means and Probability
Week 4		Density Estimation
Week 5		Gaussian Mixture Models and Expectation Maximization
Week 6		Linear Regression
Week 7		Regularization
Week 8		Singular Value Decomposition and PCA
Week 9		Bias-Variance Trade-Off
Week 10		Logistic Regression
Week 11		Evaluation of Classification Algorithms
Week 12		Gradient Descent Basics
Week 13		Convergence of Gradient Descent
Week 14		Multi-Layer Perceptron

NOTE: Schedule is tentative and subject to change.

Policies and Professional Conduct

Course Policies

All classes will be conducted face-to-face in the designated classroom. Pop-up quizzes will be administered during class sessions and will contribute to the attendance grade. The written exam will also take place in person, while homework submissions will be collected online. All announcements related to course logistics and assignments, along with course materials and lecture slides, will be shared via the course website on LearnHub.

Professional Conduct

- Student Code of Conduct
- Academic Grievance Procedure

Students are expected to submit their own work in all exams, homework assignments, and quizzes. Quizzes are designed to assess how well students have learned the most recent material; therefore, no exchange of information is allowed. Homework assignments are intended to help students enhance their knowledge and demonstrate their skills; while it is acceptable to discuss general ideas or tips with others, students must not exchange written material, work together, or allow others to complete any part of their work. During exams, all forms of communication or information transfer are strictly prohibited. Moreover, intentional participation in dishonest activities, such as helping others cheat, completing or sharing homework for others, or providing one's work to others, will also be considered cheating. Please be aware that in past semesters, several students in computer engineering courses have been referred to the Disciplinary Committee for academic dishonesty.
